

Matheus A. S. Pessoa

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SUMMARY

Research Interests: Single-molecule biophysics; nanofluidics; electrokinetic manipulation; polymer physics; fluorescence microscopy; optical methods.

Publications: 12 published / under review; 3 manuscripts in preparation.

Intellectual Property: 1 patent application.

Mentees: 9 research students.

EDUCATION

4 degrees · 2 institutions

Ph.D., Physics, McGill University 2021–Present

Montréal, Canada

Dissertation: *Electrokinetic Confinement Device: Development and Applications*

Advisor: Walter Reisner

M.Sc., Physics, McGill University 2019–2021

Montréal, Canada

Dissertation: *Development and Testing of Antenna and Subsystems for the MIST Experiment*

Advisors: H. Cynthia Chiang and Raul Monsalve

B.Sc., Physics, Federal University of ABC (UFABC) 2015–2019

São Paulo, Brazil

Thesis: *Standing-Wave Ultrasonic Scattering and Trapping of Air Microbubbles*

Advisor: Antonio A. R. Neves

B.Sc., Science & Technology, Federal University of ABC (UFABC) 2015–2018

São Paulo, Brazil

RESEARCH EXPERIENCE

5 research appointments · academic and industry experience

Nanobiophysics Laboratory, McGill University 2021–Present

Ph.D. Researcher / Research Assistant — Walter Reisner

Dimension Genomics Inc. 2022–Present

Industry Research Collaboration — Han Cao, Michael Austin

Experimental Cosmology Laboratory, McGill University 2019–2022

Research Assistant, MIST Project — H. Cynthia Chiang, Raul Monsalve

Acoustic Force and Optical Trapping Laboratory, UFABC 2018–2019

Research Assistant — Antonio A. R. Neves

Ultra-High Energy Cosmic Rays Laboratory, UFABC 2017–2018

Research Assistant — Marcelo Leigui

PUBLICATIONS

8 lead-author works · 7 co-authored works

Lead-Author Publications

1. M.A.S. Pessôa et al., *Single Molecule Capture, Release and Dynamical Manipulation via Reversible Electrokinetic Confinement (RECON)* — 2025, **Science Advances** 11(38).
2. Preethi Ravikumar & M.A.S. Pessôa et al. (equal contribution), *Electrokinetic active grooves for liposome capture, confinement, trajectory analysis and controlled release* — 2026, under review at **Biomicrofluidics**.
3. M.A.S. Pessôa (in prep) et al., *Non-equilibrium cycling conformation of molecules under electrokinetic confinement*.
4. M.A.S. Pessôa et al. (review paper in prep), *Programmable Confinement Landscapes for Single-Molecule Control in Nanofluidic Systems: From Static Nanotopographies to Reversible Electrokinetic Trapping*.
5. M.A.S. Pessôa et al., *Assessing understanding of uncertainty using the CDPA in undergraduate physics labs at McGill University* — 2026, under review at **Canadian Journal of Physics**.
6. M.A.S. Pessôa and A.A.R. Neves, *Acoustic scattering and forces on an arbitrarily sized fluid sphere by a general acoustic field* — 2020, **Journal of Sound and Vibration** 479, 115373.
7. M.A.S. Pessôa et al., *Experimental validation of the brachistochrone curve* — 2024, **Physics Education** 59(3), 035008.
8. M.A.S. Pessôa et al., *The Physics of a Disinfectant with UV-C Radiation* — 2021, **Revista Brasileira de Ensino de Física** 43.

Co-Authored Publications

1. Incl. M. Pessôa, *DNA Dynamics in Dual-Nanopore Tug-of-War* — 2026, **Macromolecules** 59(7), 4268–4278.
2. Sajad Shiekh, incl. M.A.S. Pessôa, *Proteoliposomes on 2D-MoS₂ plasmonic nanocavities for enhanced Raman spectroscopy with machine learning-based identification and classification* — 2025, **npj 2D Materials and Applications**.
3. Zelan Wu et al, (Incl. M. Pessôa), *When a Corrugated Track Beats a Straight Incline: A Scaling Law for the Optimal Corrugation* — 2026, *Submitted to European Journal of Physics*.
4. Hosseini et al, (Incl. M. Pessôa), *Tunable nanofluidic device for digital nucleic acid analysis* — 2024, **Nanoscale** 16.
5. Monsalve et al (Incl. M. Pessôa), *Mapper of the IGM spin temperature: instrument overview* — 2024, **Monthly Notices of the Royal Astronomical Society** 530.
6. H. Ferreira, M. Pessoa, et al., *The screaming balloon: predicting the sound produced by a hex nut inside a balloon* — 2019, **Emergent Scientist** 3, 7.
7. Incl. M. Pessôa, P.R. Sardelich, et al., *Method to mix tea with honey and use of transient light attenuation as an indicator of homogeneity* — 2018, **Emergent Scientist** 2, 2.

PATENT

1 patent application

Devices and Methods for Manipulating Macromolecules with an Electrokinetic Force — under review.

Matheus Pessôa, Walter Reisner, Sara Mahshid, Zezhou Liu, Imman Hosseini, Michael Austin, Han Cao.

PRIZES

4 prizes

Best Poster Award, Methods and Applications in Fluorescence Conference	2025
Innovation Prize COVID-19, Fleury Laboratories Group — 2nd Place	2020
Third Place, X International Physicists' Tournament, MIPT, Moscow	2018
Institutional Honours for Third Place at the International Physicists' Tournament, UFABC	2018

FELLOWSHIPS & AWARDS

8 fellowships and awards

FRQNT Doctoral Scholarship (CAD\$96,000)	2022–2026
Science Education Fellowship, McGill University	2020–2026
Travel Award, Biophysical Society of Canada	2026
Ambassador Award, Science Undergraduate Society, McGill University (CAD\$37,000)	2023–2025
Graduate Travel Award, McGill University (CAD\$5,000)	2020–2025
Graduate Excellence Award, McGill University	2019–2026
Trottier McGill Space Institute Graduate Award (CAD\$4,000)	2019–2021
Undergraduate Research Award, Federal University of ABC	2017, 2019

CONFERENCE TALKS

15 conference presentations · 11 talks · 4 posters

Single-Molecule Sensors and NanoSystems (S3IC), Rome <i>Single-molecule dynamical manipulation in real-time via Reversible Electrokinetic CONfinement (RECON)</i>	2026
Biophysical Society of Canada, Winnipeg <i>Harnessing polymer anisotropy through electrokinetic confinement (poster)</i>	2026
International Workshop on Disordered Systems (IWDS26), McGill University <i>Physics of electrokinetic confinement (poster)</i>	2026
Global Physics Summit – Division of Polymer Physics, Colorado <i>Reversible Electrokinetic Confinement (RECON) Device for Dynamic Confinement of Single Macromolecules and Nanoparticles</i>	2026
Methods and Applications in Fluorescence, Montréal <i>Modulated dynamics of fluorescently labeled dsDNA under electrokinetically-driven confinement (poster)</i>	2025
Global Physics Summit – Division of Polymer Physics, Anaheim <i>Modulated dynamics of single molecules under electrokinetically-driven confinement</i>	2025
Global Physics Summit – Physics Education Research, Anaheim <i>Assessing Students' Understanding of Uncertainty in Undergraduate Physics Laboratory Courses at a Major Canadian University</i>	2025
APS March Meeting – Division of Polymer Physics, Minneapolis <i>Physics of single-molecule electrokinetically driven confinement</i>	2024
Biophysical Society of Canada, Montréal <i>Dynamics of single molecules under electrokinetic confinement (poster)</i>	2024

SALTISE, Montréal <i>Analyzing and Addressing Students' Conceptual Understanding of Uncertainty in Experimental Physics Labs</i>	2024
SALTISE, Montréal <i>Assessing understanding of uncertainty using the CDPA in undergraduate physics labs at McGill University</i>	2023
Canadian Association of Physicists Congress, Ottawa <i>Integrating Group Discussion and Inquiry-Guided Learning into Physics TA Training, contributed talk</i>	2022
IEEE AP-S / URSI-GASS, Rome <i>Electrical Characterization of the Soil for Sky Radio Measurements with the MIST Global 21-cm Experiment</i>	2021
Science at Low Frequencies VII, University of Amsterdam <i>Development of the MIST antenna</i>	2020
3rd 21cm Global Signal Workshop, University of Cambridge <i>Antenna design for MIST</i>	2020

INVITED TALKS

10 invited talks and seminars

Biological Physics Lab, Federal University of Viçosa, Brazil <i>Introducing Reversible Electrokinetic Confinement Device</i>	2026
Center for Physics of Materials Student Seminar, McGill University <i>Harnessing polymer shape using electrokinetic confinement</i>	2026
Max Prigozhin Group, Harvard University <i>Introducing Reversible Electrokinetic Confinement Device</i>	2025
Soft Matter Physics, California State University, Long Beach <i>Guest Lecture, Soft Matter Physics</i>	2025
Center for Physics of Materials Student Seminar, McGill University <i>Reversible Electrokinetic Confinement</i>	2025
Office of Science Education Breakfast Club, McGill University <i>Educational initiative by the Office of Science Education (OSE)</i>	2025
Wiseman Lab, McGill University <i>Introducing Reversible Electrokinetic Confinement Device</i>	2024
Física ao Vivo, Brazilian Physical Society <i>O que é o "Brazilian Physicists' Tournament"?</i>	2023
McGill SKILLSETS Series <i>Navigating Graduate Supervision</i>	2021
McGill Space Institute Lunch Talk Series <i>What is the International Physicists' Tournament?</i>	2021

MEDIA CITATIONS & INTERVIEWS

4 selected media citations

Phys.org / McGill University Physicists maneuver DNA molecules using electrical fields, offering real-time control	Oct. 2025
McGill University Newsroom Des physiciens de McGill manipulent des molécules d'ADN au moyen de champs électriques	Oct. 2, 2025
GenomeWeb Researchers develop technology for precision control of single molecules	2025
SciAdvances Brasil Pesquisa liderada por brasileiro consegue manipular moléculas de DNA usando campos elétricos	2025 · Português

TEACHING

6 courses · 9 course-term appointments · 2 education/coordination roles

Teaching Experience, Department of Physics, McGill University

Physics Olympiad Coordinator Department of Physics, McGill University	F2024, W2025
Advanced Lab Modern Physics Teaching Assistant	F2025, W2026
Experimental Methods I Teaching Assistant	F2019, F2023
Experimental Methods II Teaching Assistant	W2020, W2021
Introductory Mechanics Grader and Tutorial Lead	F2021
Introductory Electricity & Magnetism Grader and Tutorial Lead	W2022
Physics of Music Grader and Tutorial Lead	F2020

TEACHING LEADERSHIP & TRAINING

Teaching Assistant Training in Physics — Developer and leader	2020–2026
Teaching Assistant Training in Chemistry — Facilitator	2020–2026
Teaching Assistant Training in Computer Science — Facilitator	2020–2026

ACADEMIC LEADERSHIP & SERVICE

7 leadership and service roles

Physics Education Research Group — Leader	2020–2026
Undergraduate Research Showcase — Organizer	2021–2026
Center for Physics of Materials Student Committee, McGill University	2025–2026
McGill Graduate Association of Physics Students — TA Officer	2022–2023
McGill Physics Hackathon — Mentor	2019–2020
Antenna Development Lead, McGill Rocket Team — Mentor	2019–2020
Lunch with the Seminar Speaker, McGill Space Institute — Organizer	2019–2020

ACADEMIC COMPETITION LEADERSHIP

5 competition leadership roles

International Physicists' Tournament — Executive Committee / Organizer	2017–Present
Canadian International Physicists' Tournament — Founder / National Representative	2021–Present
International Physicists' Tournament — Juror	2020–Present
International Young Physicists' Tournament Brazil — Organizer	2017–2018
Brazilian International Physicists' Tournament — Founder / National Representative	2015–2019

RESEARCH MENTORSHIP

9 research mentees · 2 thesis projects

Preethi Ravikumar (PhD thesis)	2024–Present
Seraphine Kautz	May 2024–Present
Sarah Ameer	May–Sep 2024
Piotr Jakuc	May–Sep 2023
Carolina Queiroz	May–Sep 2023
Naomi Duggan (BSc thesis)	Apr 2023
Jacob Kennedy	2020
Aiky Rasolomanana	2020
Megan Cvitan	2019

OUTREACH

6 outreach initiatives

Montreal Young Physicists Expo — Organizer	2026
Summer School of Science (S3), Zagreb — Project Leader	2023
Low-Cost Intelligent UV-C Sterilizer Project, UFABC	2020
Science Communication podcast in Portuguese: Física Popular Brasileira	2019–Present
How Physics Can Change the World, UFABC	2017–2018
Enactus UFABC — Project Leader	2015–2019

SCIENCE COMMUNICATION

2 published articles

- M.A.S. Pessoa et al., *Já pensou na existência de um esterilizador inteligente utilizando radiação ultravioleta-C? Conheça o BURN.* — 2020, Blog Divulga Ciência, UFABC.
- M.A.S. Pessoa et al., *BURN — Esterilizador Inteligente de Baixo Custo Utilizando Radiação Ultravioleta-C* — 2020, Revista Pesquisa ABC.

TECHNICAL SKILLS

Programming · simulation · microscopy · nanofabrication · biomolecular preparation · design

Programming: Python, MATLAB, Mathematica, LaTeX.

Simulation & Analysis: ESPResSo, COMSOL, FlexPDE, ImageJ/Fiji, Micro-Manager, Ilastik.

Microscopy & Instrumentation: Widefield fluorescence microscopy; single-molecule imaging; RF and electrical instrumentation.

Micro/Nanofabrication: Lithography; mask design; ALD; PECVD; sputtering; chemical and reactive-ion etching; 3D printing.

Biomolecular Preparation: Fluorescent labeling of DNA, liposomes, and plasmids.

Design: CAD; KLayout; experimental prototyping.

LANGUAGES

6 languages

Portuguese (native); English (C2); Spanish (B2); French (B2); Russian (A2); German (A2).